

American science needs an executive branch constrained by checks and balances



In the USA, physics research is suffering from new and proposed government policies that divert funds away from research, politicize funding decisions and make it harder to collaborate or engage in the international community. These harmful actions are not the result of Congress, which has largely continued to support federal funding of research at similar levels as before, but result from a newly emboldened Executive Branch that acts unconstrained by checks and balances and subverts the will of Congress and the courts. This threat to research will require new strategies for advocacy that reassert the importance of checks and balances on the executive.

Among the most damaging positions taken by the Executive Branch is its inability or unwillingness to disburse research funds provided by Congress. Despite steady congressional appropriations, the National Science Foundation (NSF) has so far this year awarded **40% fewer grants** than they had awarded this time in the last fiscal year. This is in defiance of a Congressional directive that explicitly forbids the NSF from dramatic reductions in funding¹. The Executive Branch has already shown its willingness to either ignore or slow walk corrective action by the courts in related cases. For example, in January 2025 the Executive Branch froze federal research funding across many agencies, including at the Department of Energy (DOE), and then **ignored a court order** to reinstate them. A court ruled this “**likely unconstitutional**”.

The courts have already frequently found that the Executive Branch acted unlawfully in implementing policies that harmed scientists. A new US\$100,000 fee on H1-B visas has made it more challenging to bring international scientists to the USA, and **the courts ruled this unconstitutional**, because only Congress has the power to set that fee. Executive action to gut the National Center for Atmospheric Research by giving away their supercomputers was recently blocked by the courts², and the mass firing of probationary workers at NSF was also eventually ruled unlawful³.

The Executive Branch is still pursuing harmful policies even when it is clear that they have

little chance of survival in court. For example, the DOE unilaterally cut overhead rates despite a court ruling forbidding an identical earlier move by the National Institutes of Health⁴, and **backed down only when Congress stepped in**. Reported NSF blacklisting of specific universities⁵ is another example of increased Executive Branch power that is unlikely to survive court challenge but is certainly doing harm in the meantime.

Perhaps most worrisome is a set of proposed rules by the Office of Management and Budget that would **curtail international collaborations and diminish the role of peer review in federal science funding, giving political appointees the authority to make funding decisions for individual grants**, including on the basis of non-scientific considerations. If such devastating dramatic changes are to be made, they should come from Congress, not the Executive Branch.

For the physics community, these new attacks suggest that our usual advocacy playbook needs to change. In the past it has been sufficient to lobby Congress to fund research but now that is insufficient because the root cause of many of the challenges facing scientists in the USA do not stem from a lack of support of Congress. Instead, they come from an Executive Branch that no longer feels beholden to carry out Congress’ laws. An Executive Branch that is subject to constraints by Congress and the courts is better for stability and for science.

Professional societies also need to speak up for a constrained executive. Scientists can help by contacting their professional societies to encourage them to take a stronger, more vocal stance against executive overreach. Similarly, we can call our congressional members to demand more oversight to ensure that science is funded according to Congress’ will.

For scientists outside the USA, this is a good time to think about how to prevent or push back against similar emboldened executive action in countries that might face such challenges after future elections. In particular, the USA experience has taught us that it is essential to build, in advance, strong alliances among professional societies as well as universities

and other research institutions. The purpose of such alliances would be to respond to future challenges collectively, instead of separately.

These conditions require us to step outside of our comfort zones. As a scientist, it can be tempting to think of one’s professional self as above politics, and it is easy to fall back on old adages that suggest we should stick to the science. But this would be a mistake. We now confront an unconstrained executive that is **changing how policy and politics is conducted in the USA**. Failing to engage now will leave us with little science to which we can stick. It is time for all citizens, especially those interested in the betterment of society through science, to stand up for the checks and balances guaranteed by the US Constitution.

Note: As this article went to press, the DOE stated that in 2025 it **cancelled grants based solely on whether the recipients were in states that voted against the president**. A trial court ruled that **such cancellations would be unconstitutional**, but the administration’s proposed rule changes would seem to codify this practice.

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Competing interests

The authors declare no competing interests.